STM32F407V

Recommended first-candidate mapping for STM32F407VG (LQFP100). Assignments are grouped t

No.	Section	Function / Signal	STM32 Pin	Peripheral / Mode
1	Power Sensing	R Phase Voltage	**PC0**	ADC1_IN10
2	Power Sensing	Y Phase Voltage	**PC1**	ADC1_IN11
3	Power Sensing	B Phase Voltage	**PC2**	ADC1_IN12
4	Power Sensing	R Phase Current	**PC3**	ADC1_IN13
5	Power Sensing	Y Phase Current	**PC4**	ADC1_IN14
6	Power Sensing	B Phase Current	**PC5**	ADC1_IN15
7	Control Pilot	CP Voltage Sense	**PB0**	ADC1_IN8
8	Temperature	Analog Temperature Sensor	**PB1**	ADC1_IN9
9	Control Pilot	CP PWM	**PE9**	TIM1_CH1
10	Safety	RCM Fault	**PE0**	GPIO Input + EXTI0
11	Power Control	Contactor 1 Feedback	**PE1**	GPIO Input
12	Power Control	Contactor 2 Feedback	**PE2**	GPIO Input
13	Power Control	Contactor 1 Control	**PE3**	GPIO Output
14	Power Control	Contactor 2 Control	**PE4**	GPIO Output
15	Communication	UART TX	**PD8**	USART3_TX
16	Communication	UART RX	**PD9**	USART3_RX
17	RFID	RFID CS / NSS	**PA4**	GPIO Output (SPI1_NSS capable)
18	RFID	RFID SCK	**PA5**	SPI1_SCK
19	RFID	RFID MISO	**PA6**	SPI1_MISO
20	RFID	RFID MOSI	**PA7**	SPI1_MOSI
21	User Interface	Buzzer	**PD10**	GPIO Output
22	Display	16×2 LCD RS	**PD0**	GPIO Output
23	Display	16×2 LCD EN	**PD1**	GPIO Output
24	Display	16×2 LCD D4	**PD2**	GPIO Output
25	Display	16×2 LCD D5	**PD3**	GPIO Output
26	Display	16×2 LCD D6	**PD4**	GPIO Output
27	Display	16×2 LCD D7	**PD5**	GPIO Output
28	Debug	SWDIO	**PA13**	SWDIO
29	Debug	SWCLK	**PA14**	SWCLK

/G EV Charger – Recommended

:o keep analog sensing, SPI, UART, LCD, control

Direction	Port Group	Priority / Constraint
Input	GPIOC	High – ADC pin
Input	GPIOC	High – ADC pin
Input	GPIOC	High – ADC pin
Input	GPIOC	High – ADC pin
Input	GPIOC	High – ADC pin
Input	GPIOC	High – ADC pin
Input	GPIOB	High – ADC pin
Input	GPIOB	High – ADC pin
Output	GPIOE	High – timer AF
Input	GPIOE	High – safety interrupt
Input	GPIOE	Normal GPIO
Input	GPIOE	Normal GPIO
Output	GPIOE	Normal GPIO
Output	GPIOE	Normal GPIO
Output	GPIOD	Peripheral AF
Input	GPIOD	Peripheral AF
Output	GPIOA	SPI group
Output	GPIOA	SPI AF
Input	GPIOA	SPI AF
Output	GPIOA	SPI AF
Output	GPIOD	Normal GPIO
Output	GPIOD	Normal GPIO
Output	GPIOD	Normal GPIO
Output	GPIOD	Normal GPIO
Output	GPIOD	Normal GPIO
Output	GPIOD	Normal GPIO
Output	GPIOD	Normal GPIO
Bidirectional	GPIOA	Reserved
Input	GPIOA	Reserved

Pin Assignment

and debug signals organized.

Comment / Design Reason
R-phase voltage analog input. Grouped with the other three-phase sensing channels on GPIOC.
Y-phase voltage analog input. Sequential ADC channel grouping simplifies schematic and firmware
B-phase voltage analog input.
R-phase current analog input.
Y-phase current analog input. Note: PC4 is also an Ethernet RMII-related pin; acceptable because
B-phase current analog input. Note: PC5 is also an Ethernet RMII-related pin; acceptable for the
Analog Control Pilot measurement input.
Analog-output temperature sensor. Keeps the eighth analog channel adjacent to CP sensing.
Hardware timer PWM output for the Control Pilot signal. TIM1_CH1 provides accurate PWM generation.
Residual-current monitor fault input. EXTI0 allows immediate interrupt-driven fault handling.
Reads auxiliary feedback from contactor 1. Polling is sufficient unless requirements later demand an
Reads auxiliary feedback from contactor 2.
Controls the external driver circuit for contactor 1. The MCU must not drive the contactor coil directly.
Controls the external driver circuit for contactor 2.
UART transmit line using USART3. PD8/PD9 form a clean TX/RX pair.
UART receive line using the same USART3 peripheral as TX.
Software-controlled RFID chip select. PA4 is also SPI1_NSS capable. Do not tie CS permanently low.
SPI1 clock line for the RFID reader.
SPI1 data from RFID reader to STM32.
SPI1 data from STM32 to RFID reader. PA4–PA7 keep the complete RFID SPI interface together.
Suitable for an active buzzer. If a passive buzzer with tone control is later selected, move it to a timer
LCD Register Select. LCD is operated in 4-bit mode.
LCD Enable signal.
LCD data bit D4 in 4-bit mode.
LCD data bit D5 in 4-bit mode.
LCD data bit D6 in 4-bit mode.
LCD data bit D7 in 4-bit mode. LCD R/W can be tied to GND if the firmware only writes to the display.
Keep reserved for STM32 programming and debugging.
Keep reserved for STM32 programming and debugging.